

Bugs found: 3

1. Buffer overflow: loop iterates 4 times ($i < 4$) but array 'v' has only 3 elements (indices 0-2), writing out of bounds at `v[3]`.
2. Uninitialized variable: 'soma' is used in the accumulation loop without being initialized, causing undefined behavior.
3. Memory leak: memory allocated with `malloc()` for 'v' is never freed before the function returns.